Cudgeling our brains over possessive idioms

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DELPH-IN 2015

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Overview

- Motivation: why do we care about idioms?
- Syntax
- Semantics
  - decomposable
  - non-decomposable
- Variation and corpus processing
- Successive approximation

Idioms don't translate

(1) *Kim blew her top*

"Kim got angry"

No blowing, no top, no her

(2) キムは　頭に　来た

kimu-wa atama-ni kita

Kim-TOP head-DAT came

"Kim got angry (lit: Kim came to her head)"

no head, no coming

- They are hard to identify
- They require work to represent

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The state of the art (translation)

(3) *Kim blew her top*

キムは、 彼女の 上を 吹きました
Kimu wa, kanojo no ue o fukimashita
Kim-TOP her-'s top-ACC blew

Kim exhaled on the upper part of her (Google Translate)

キムは、 彼女の トップを 吹いた
Kimu wa, kanojo no toppu o fuita
Kim-TOP her-'s top-ACC blew

Kim exhaled on the shirt of her (Bing Translate)

(4) キムは 頭に 来た

Kim came to the head (Google+Bing Translate)

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The state of the art (wordnet)

- Hyponym of *rage* with many lemmas
  - *flip one's lid*
  - *blow one's stack*
  - *flip one's wig*
  - *lose one's temper*
  - *blow up*
  - *throw a fit*
  - *hit the roof*
  - *hit the ceiling*
  - *have kittens*
  - *have a fit*
  - *combust*
  - *fly off the handle*
  - *blow a fuse*
  - *go ballistic*
- But by no means exhaustive
  - *lose one's cool*
  - *lose one's rag*
  - *lose one's grip*
  - *blow one's top*
  - *fly off one's handle*
  - *spit the dummy*

Possessed Idioms (i)

Structure	Example	Frequency
X_{NP} V_1 X's N_1	*lose one's mind*	137
X_{NP} V_1 [P_1 X's N_1]	*fly off one's handle*	40
X_{NP} V_1 X's N_1 [P_1 Y_{NP}]	*cast one's lot with someone/thing*	39
X_{NP} V_1 X's N_1 [P_1 D_1 N_2]	*have one's head in the clouds*	27
X_{NP} V_1 X's N_1 P_1	*cry one's eyes out*	22
X_{NP} V_1 X's own N_1	*blow one's own horn*	18
X_{NP} V_1+P_1 X's N_1	*pull up one's socks*	17
X_{NP} be [P_1 X's N_1]	*[be] off one's rocker*	13
X_{NP} V_1 X's N_1 [P_1 X's N_2]	*scratch one's ear with one's elbow*	13
X_{NP} V_1 D_1 N_1 [P_1 X's N_2]	*a dose of one's medicine*	10
X_{NP} V_1 X's N_1 A_1	*get one's hands dirty*	10
X_{NP} V_1 Y_{NP} [P_1 X's N_1]	*wind someone around one's finger*	10

- Extracted from lexicons
- Collected by me and my students

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Possessed idioms (ii)

X_{NP} V_1 X's A_1(est)	*do one's best*	8
X_{NP} V_1 [P_1 X's N_1 [P_2 Y_{NP}]]	*pour out one's heart to someone*	7
X_{NP} aux+neg V_1 X's N_1	*not mince one's words*	5
X_{NP} V_1 Y_{NP} D_1 N_1 [P_1 X's N_2]	*give someone a piece of one's mind*	4
X_{NP} V_1 R_1 A_1 [P_1 X's N_1]	*too big for one's boots*	3
X_{NP} V_1 [P_1 D_1 N_1 P_2 X's N_2]	*by the skin of one's teeth*	2
X_{NP} V_1 N_1 [P_1 X's N_2]	*have egg on one's face*	2
X_{NP} V_1 X's N_1 [P_1 X]	*have one's wits about one*	2
X_{NP} V_1 X's N_1 and V_2 N_2	*have one's cake and eat it*	2
Remainder	*let grass grow under one's feet*	30
Total		**421**

- Some of these are NPI:

(5) *Kim doesn't let the grass grow under his feet*

(6) ?*Kim lets the grass grow under his feet*

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The idiom lexicon

(7)

Idiom entry	
Index form	eat one's words
Template	X_{NP} V_1 X's N_1
Example	Kim eats her words
Example	Kim is going to have to eat her words
Definition	to retract one's statement, especially with humility
V_1	S: (v) eat (take in solid food)
N_1	S: (n) words (the words that are spoken)
$^*V_1$	S: (v) swallow, take back, unsay, withdraw (take back what one has said)
$^*N_1$	S: (n) statement (a message that is stated or declared; a communication (oral or written) setting forth particulars or facts etc)
. . .	

Idiom entry ...	
Index form	eat one's words
@type	decomposable
Paraphrase	X retracts X's statement
@template	X V N
@V	S: (v) abjure, recant, forswear, retract, resile (formally reject or disavow a formerly held belief, usually under pressure)
@N	S: (n) statement (a message that is stated or declared; a communication (oral or written) setting forth particulars or facts etc)

(8)

Idiom entry	
Index form	twiddle one's thumbs
Template	X_{NP} V_1 X's N_1
Example	Kim twiddles her thumbs
Definition	to do nothing
V_1	S: (v) twiddle, fiddle with (manipulate, as in a nervous or unconscious manner)
N_1	S: (n) thumb, pollex (the thick short innermost digit of the forelimb)
@type	Nondecomposable
Paraphrase	X is idle
@template	X BE A
@A	S: (adj) idle (not in action or at work))

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How do we recognize them?

(9) *Kim$_i$ gets^I her$_i$ break*. [X_{NP} V_1 X's N_1]

(10) Idiom filter rule:

$$\begin{bmatrix} \text{get_i_rel} & (X_i, \text{break_rel}) \\ \text{poss_rel} & (\text{pron_rel}, \text{break_rel}) \\ \text{id_rel} & (X_i, \text{pron_rel}) \end{bmatrix}$$

- `id_rel` enforces agreement between target and pronoun.
- Every parse with an idiomatic predicate ([IDIOM +]) must be licensed by an idiom rule, or the parse is discarded.
- This check is done when checking the root condition.
- The idiom rules flexibly identify the idioms.

The AVM

(11)

$$
\begin{bmatrix}
\textit{mrs} \\
\text{TOP} & \boxed{0}\ h \\
\text{INDEX} & \boxed{2}\ e \\
\text{RELS} & \left\langle
\begin{bmatrix} \textit{proper_q_rel} \\ \text{LBL} & \boxed{4}\ h \\ \text{ARG0} & \boxed{3}\ x \\ \text{RSTR} & \boxed{5}\ h \\ \text{BODY} & \boxed{6}\ h \end{bmatrix},
\begin{bmatrix} \textit{named_rel} \\ \text{LBL} & \boxed{7}\ h \\ \text{CARG} & \textit{"kim"} \\ \text{ARG0} & \boxed{3}\ x \end{bmatrix},
\begin{bmatrix} \textit{_get_v_i_rel} \\ \text{LBL} & \boxed{1}\ h \\ \text{ARG0} & \boxed{2}\ e \\ \text{ARG1} & \boxed{3}\ x \\ \text{ARG2} & \boxed{9}\ x \end{bmatrix},
\begin{bmatrix} \textit{id_rel} \\ \text{LBL} & \boxed{1}\ h \\ \text{ARG0} & \boxed{10}\ i \\ \text{ARG1} & \boxed{3}\ x \\ \text{ARG2} & \boxed{11}\ x \end{bmatrix}, \right. \\
& \left.
\begin{bmatrix} \textit{poss_rel} \\ \text{LBL} & \boxed{15}\ h \\ \text{ARG0} & \boxed{16}\ e \\ \text{ARG1} & \boxed{9}\ x \\ \text{ARG2} & \boxed{11}\ x \end{bmatrix},
\begin{bmatrix} \textit{pronoun_q_rel} \\ \text{LBL} & \boxed{17}\ h \\ \text{ARG0} & \boxed{11}\ x \\ \text{RSTR} & \boxed{18}\ h \\ \text{BODY} & \boxed{19}\ h \end{bmatrix},
\begin{bmatrix} \textit{pron_rel} \\ \text{LBL} & \boxed{20}\ h \\ \text{ARG0} & \boxed{11}\ x \end{bmatrix}, \right\rangle \\
& \begin{bmatrix} \textit{def_explicit_q_rel} \\ \text{LBL} & \boxed{12}\ h \\ \text{ARG0} & \boxed{9}\ x \\ \text{RSTR} & \boxed{13}\ h \\ \text{BODY} & \boxed{14}\ h \end{bmatrix},
\begin{bmatrix} \textit{_break_n_1_rel} \\ \text{LBL} & \boxed{15}\ h \\ \text{ARG0} & \boxed{9}\ x \end{bmatrix} \\
\text{HCONS} & \left\langle
\begin{bmatrix} \textit{qeq} \\ \text{HARG} & \boxed{0}\ h \\ \text{LARG} & \boxed{1}\ h \end{bmatrix},
\begin{bmatrix} \textit{qeq} \\ \text{HARG} & \boxed{5}\ h \\ \text{LARG} & \boxed{7}\ h \end{bmatrix},
\begin{bmatrix} \textit{qeq} \\ \text{HARG} & \boxed{13}\ h \\ \text{LARG} & \boxed{15}\ h \end{bmatrix},
\begin{bmatrix} \textit{qeq} \\ \text{HARG} & \boxed{18}\ h \\ \text{LARG} & \boxed{20}\ h \end{bmatrix} \right\rangle \\
\text{ICONS} & \langle\ \rangle
\end{bmatrix}
$$

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Semantics

- Tweak the meaning a little.
  - If the structure can be kept (**decomposable**) then map the predicates to appropriate wordnet senses.

    (12) *Kim changes his tune* "Kim changed his opinion"
  - If it is more opaque (**non-decomposable**) then rewrite to a more suitable semantics.

    (13) *Kim blows her mind* "Kim becomes angry"
- Jia Qian mapped predicates to wordnet and wrote paraphrases for all of the idioms!
- But they have not all been put in the ERG (yet).
  - We can use the translation machinery to do this

Quantum Superpositioning?

- The easiest thing to do is just add a new layer
  - MRS $\rightarrow$ paraphrased MRS
- *"Stop worrying. You wear your heart on your sleeve, and your face."* Coulson to Skye (*Agents of S.H.I.E.L.D.* "Trust no one, suspect everyone")
  - So we really want to access both layers (idiomatic and surface) at once
  - We can pack the MRSs (expecially DMRs) **conjunctively** not **disjunctively**
  - This requires machinery we don't yet have

Frequency

	Decomposable		Nondecomposable		Total
Co-indexed possessive idioms	159	(37.8%)	262	(62.2%)	421
Separate possessive idioms	32	(34.4%)	61	(65.6%)	93
Total number of idioms	191	(37.2%)	323	(62.8%)	514

- No correlation with transparency (is the idiomatic meaning obviously motivated) or plausability (can this actually happen)

Corpus-based study

- Currently running over the BNC to find examples
- All idioms expanded to variants using synonyms and hyponyms of the literal meaning of the open class words.
  blow one's top $\rightarrow$ *exhale one's head*, . . .
- Pilot study with just X_{NP} V_1 X_{NP}'s N_1
- This type found in 0.03% of sentences
- Almost no examples found of variation that is not already in the lexicon.

Examples of modification in the BNC

(14) *cannot believe my own bloody eyes*

(15) *make one's unsteady way*

(16) *have one's humorous moment*.

Future Work

- Release the idiom database
  - Ask people for more entries
- Add everything to the ERG/wordnet
- Re-parse all our corpora
- Tie up some loose threads
  - modification (quantum MRS)
  - variation
  - alternation
- Look at more classes

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Alternation in Idioms

S/N	Cause-type Idiom	Result-type Idiom
1	*give someone someone's dues* X_{NP} V_1 Y_{NP} Y's N_1	*get one's dues* Y_{NP} V_1 Y's N_1
2	*give someone someone's walking papers* X_{NP} V_1 Y_{NP} Y's N_1	*get one's walking papers* Y_{NP} V_1 Y's N_1
3	*give someone a dose/taste of someone's own medicine* X_{NP} V_1 Y_{NP} D_1 N_1 [P_1 Y's N_1]	*get/receive/have/need a dose/taste of one's own medicine* Y_{NP} V_1 D_1 N_1 [P_1 Y's N_2]
4	*give someone a run for someone's money* X_{NP} V_1 Y_{NP} D_1 N_1 [P_1 Y's N_1]	*get/have a run for one's money* Y_{NP} V_1 D_1 N_1 [P_1 Y's N_2]
5	*give someone a piece of one's mind* X_{NP} V_1 Y_{NP} D_1 N_1 [P_1 X's N_2]	*get a piece of someone's mind* Y_{NP} V_1 D_1 N_1 [P_1 X_{NP}'s N_2]

6	*give someone the rough edge of one's tongue* X_{NP} V_1 Y_{NP} D_1 A_1 N_1 [P_1 X's N_2]	*get the rough edge/side of someone's tongue* Y_{NP} V_1 D_1 A_1 N_1 [P_1 X_{NP}'s N_2]
7	*put someone's nose out of joint* X_{NP} V_1 Y_{NP}'s N_1 [P_1 N_2]	*get/have one's nose out of joint* Y_{NP} V_1 Y's N_1 [P_1 N_2]
8	*bring someone out of someone's shell* X_{NP} V_1 Y_{NP} [P_1 Y's N_1]	*come out of one's shell* Y_{NP} V_1 [P_1 Y's N_1]
9	*get/have/take someone under one's wing* X_{NP} V_1 Y_{NP} [P_1 X's N_1]	*be/come under someone's wing* Y_{NP} V_1 [P_1 X_{NP}'s N_1]
10	*have/keep someone under one's thumb* X_{NP} V_1 Y_{NP} [P_1 X's N_1]	*be under someone's thumb* Y_{NP} V_1 [P_1 X_{NP}'s N_1]
11	*let someone/thing slip through one's fingers* X_{NP} V_1 N_1 V_2 [P_1 X's N_2]	*slip through someone's fingers* Y_{NP} V_1 X_{NP}'s N_1

The Integrated Semantic Framework

ISF Goals

- A tractable, achievable representation
- Model similar meanings through similar structure and closeness in the lexical hierarchy
- Exploit human work on encoding knowledge
  - Grammarians for structural semantics
  - Lexicographers for lexical semantics
- Monostratal (constraint-based with ML ranking)
  - Represent different levels of information in the same structure
  - ordering pipelines is hard — each module wants the other
  - "*premature optimization is the root of all evil*" (Knuth, 1974)

Consistent predicate granularity

- Some decomposition:
  - *I live here*: "in/on/at this place" `in(this(place))`
  - *codevelop*: "co- develop" `together` $\wedge$ `develop`
  - *long-legged*: "has/with long legs" `with(legs)` $\wedge$ `long(legs)`
- Many MWEs:
  - *I look it up*: "look up";
  - *Poisson and Gaussian distributions*
    $\boldsymbol{Poisson\ distribution}_{n:1}$ and $\boldsymbol{Gaussian\ distribution}_{n:1}$
  - *I wrack my brains*: "think hard"
  - *She blows her top*: "gets angry"

- Integrated Disambiguation models (ongoing work):
  - parse (re)ranking and word sense disambiguation
  - under-specify when possible; prune if necessary

The ontology is part of the model

- *destruction* is linked to "destroy" (derivation)
  - One hop away, we are fine:
    speckled band: VBN or JJ (derived from V)
- *teacher* is linked to "person who teaches" (hypernym + derivation)
  (should type the links better)
- more robust than paraphrasing
- richer links between synsets are needed

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What makes it possible now?

- Breakthroughs in robustness for constraint-based grammars
  - wider coverage of core elements
  - robust handling of unknown words
  - efficient packing
  - robust handling of unknown relations New
- Access to wordnets in multiple languages
  - 28 curated open wordnets (1 new one every 2 months)
  - not just translating the Princeton wordnet
  - $\Rightarrow$ new links, new concepts, new meta-data
- Better analyses of many phenomena

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What do we produce?

- Dependency between predicates with underspecified quantifiers and a ranking model
- Predicates linked to wordnet
  and a disambiguation model (UKB, LeLesk)
- Uniform across multiple languages
  English, Japanese, Mandarin Chinese, Indonesian/Malaysian
  (Cantonese, Abui) (Spanish, German, Norwegian, Dutch)
- Open source, incrementally released (with python modules)

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Please Join in

- Wordnet annotation/Meaning Bank for German, Dutch, Italian, . . .
- Coordinating with wordnet projects
- Will use the data to add sense-frequencies for wordnets
- Annotating *Speckled Band* in a new language is a perfect size for an undergraduate thesis
  - We will make our software available to do this
- We can project sentences from our current annotation
  - English, Chinese, Japanese (Indonesian under way)

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Thanks and more

- We would like to thank:
  - Erasmus Mundus Multi and Elke for bringing me here
  - Global Wordnet Association (GWA)
  - Deep Linguistic Processing with HPSG Initiative (DELPH-IN)
  - The Creative Commons Catalyst Grant: *Assessing the effect of license choice on the use of lexical resources*
  - The JSPS-NTU grant: *Revealing Meaning Using Multiple Languages*
  - The NTU Tier 1 grant: *Shifted in Translation*
  - The MOE Tier 2 grant: *That's what you meant: A Rich Representation for Representing Meaning*
  - NTU URECA, Final Year Projects and Student Annotators
  - NTT, NICT, Google, Fuji-Xerox